1/18/23

§ 1.2 (continued)

Diagonalizability

If $A \in M_n$ is similar to a diagonal matrix $\left(\begin{bmatrix} d_1 & & & 0 \\ & d_2 & & \\ & & \ddots & \\ 0 & & & d_n \end{bmatrix} \right)$ we call A diagonalizable.

i.e. A diagonalizable

$$A = S \Lambda S^{-1}, \quad \Lambda = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$$

$$\sigma(\Lambda) = \{\lambda_1, \lambda_2, \ldots, \lambda_n\} = \sigma(A)$$

Q: How do we recognize & achieve diagonalizability?

<u>Theorem</u> $A \in M_n$ is diagonalizable if and only if there exist a set of n linearly independent eigenvectors of A.

(i.e. A is diag/ble iff $\exists$ basis for $\mathbb{C}^n$ comprising eigenvectors of A)

Proof

($\Leftarrow$) Assume $A x^{(i)} = \lambda_i x^{(i)}$, $x^{(i)} \neq 0$,

$\lambda_1, \lambda_2, \ldots, \lambda_n \in \sigma(A)$, $\{x^{(1)}, x^{(2)}, \ldots, x^{(n)}\}$

is linearly independent. Take

$$S = \left[x^{(1)} \middle| x^{(2)} \middle| \cdots \middle| x^{(n)} \right]$$ ← matrix in M_n whose columns are the eigenvectors

S^{-1} exists! Then

$$AS = A\left[x^{(1)} \middle| x^{(2)} \middle| \cdots \middle| x^{(n)}\right] =$$

$$= \left[Ax^{(1)} \middle| Ax^{(2)} \middle| \cdots \middle| Ax^{(n)} \right]$$

$$= \left[\lambda_1 x^{(1)} \middle| \lambda_2 x^{(2)} \middle| \cdots \middle| \lambda_n x^{(n)} \right]$$

$$\overset{!}{=} \left[x^{(1)} \middle| x^{(2)} \middle| \cdots \middle| x^{(n)} \right] \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$$

$$= S\Lambda, \quad \Lambda = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$$

$$\Longrightarrow \quad S^{-1}AS = S^{-1}S\Lambda = \Lambda$$

$$\Longrightarrow \quad \boxed{A = S\Lambda S^{-1}}$$ diagonalizable.

($\Rightarrow$) If $A = S\Lambda S^{-1}$, $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$,

then $AS = S\Lambda$. So if

$S = [x^{(1)} | x^{(2)} | \dots | x^{(n)}]$, then

$AS = S\Lambda \Rightarrow Ax^{(i)} = \lambda_i x^{(i)}$

& $\{x^{(1)}, x^{(2)}, \dots, x^{(n)}\}$ are lin. independent

as columns of an invertible S;

so they are eigenvectors ■

<u>Counterexamples</u>

(i) Not all matrices are diagonalizable.

$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\sigma(A) = \{0, 0\}$.

If A were diagonalizable, then

$\exists$ invertible S such that

$S^{-1}AS = \Lambda = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

$\Rightarrow A = S\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}S^{-1} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

contradiction, since $A \neq \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

(2) Not all irreducible matrices are diagonalizable.

eg $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $\sigma(A) = \{1, 1\}$

if $S^{-1}AS = \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then

$$A = S\underbrace{\Lambda}S^{-1} = SIS^{-1} = SS^{-1} = I$$

contradiction $A \neq I$.

Since previous theorem has a condition that is hard to detect, let us look for "sufficient" conditions of diag/lity.

<u>Proposition</u> Let $x^{(1)}, x^{(2)}, \ldots, x^{(k)}$ be eigenvectors of $A \in M_n$ that correspond to distinct eigenvalues $\lambda_1, \lambda_2, \ldots \lambda_k$, respectively. Then

$$\{x^{(1)}, x^{(2)}, \ldots, x^{(k)}\} \quad (k \leq n)$$

is a linearly independent set.

Proof

<u>Proof</u> We have $Ax^{(i)} = \lambda_i x^{(i)}$, $i = 1, 2, \ldots, k$,

$\lambda_i \neq \lambda_j$ if $i \neq j$, $x^{(i)} \neq 0$.

Say, by way of contradiction,

$$\boxed{\sum_{i=1}^{r} \alpha_i x^{(i)} = 0} \quad (1)$$

for some $\alpha_i \in \mathbb{C}$

where r = minimum number possible among all linear combinations of the $x^{(i)}$'s with all coefficients $\alpha_i \neq 0$. $(r \leq k)$

Now by (1)

$$0 = A\left(\sum_{i=1}^{r} \alpha_i x^{(i)}\right)$$

$$= \sum_{i=1}^{r} \alpha_i A x^{(i)}$$

$$\boxed{0 = \sum_{i=1}^{r} \alpha_i \lambda_i x^{(i)}} \quad (2)$$

Compute (2) $- \lambda_r \times$ (1)

$$\sum_{i=1}^{r} \alpha_i (\lambda_i - \lambda_r) x^{(i)} = 0$$

$$\Rightarrow \sum_{i=1}^{r-1} \underbrace{\alpha_i}_{\neq 0} \underbrace{(\lambda_i - \lambda_r)}_{\neq 0} x^{(i)} = 0$$

Contradiction to minimality of r !!!

This contradicts essentially (1), namely that the eigenvectors are linearly dependent.

So $\{x^{(1)}, x^{(2)}, \dots, x^{(k)}\}$ are linearly independent ∎

$$\alpha^{(1)} x^{(1)} + \alpha^{(2)} x^{(2)} = 0$$

$$- \quad \lambda_2 \alpha^{(1)} x^{(1)} + \lambda_2 \alpha^{(2)} x^{(2)} = 0 \quad \boxed{\lambda_2 \otimes (1)}$$

$$+ \quad \lambda_1 \alpha_1 x^{(1)} + \lambda_2 \alpha^{(2)} x^{(2)} = 0 \quad (2)$$

$$- \lambda_2 \alpha_1 x_1^{(1)} + \lambda_1 \alpha_1 x^{(1)} = 0$$

$$\underbrace{(\lambda_2 - \lambda_1)}_{\neq 0} \underset{\neq 0}{\alpha_1} x^{(1)} = 0$$

<u>Corollary</u> If $A \in M_n$ has n distinct eigenvalues, then A is diagonalizable.

<u>Comment</u> $\begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} = I$ is diagonalizable but eigenvalues are <u>not</u> distinct.

1/20/23

Commutativity & diagonalizability

One of the most intriguing concepts in matrix theory is that of commutativity:

$A, B \in M_n$ are <u>commuting</u> (they commute) if $AB = BA$.

Generically, $AB \neq BA$. It's a special occurrence if they commute.

How special?

<u>Lemma</u> Let $A \in M_n$, $B \in M_m$. Construct

$$C := A \oplus B = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \in M_{n+m}$$

"direct sum"

Then C is diagonalizable if and only if A, B are diagonalizable.

<u>Proof</u> Read the details in book <u>or</u> prove it on your own.

(in fact, $\sigma(C) = \sigma(A) \cup \sigma(B)$) ∎

<u>Definition</u> $A, B \in M_n$ are called <u>simultaneously diagonalizable</u> if there exists invertible $S \in M_n$ such that

$$S^{-1}AS, \quad S^{-1}BS$$

are both diagonal matrices.

<u>Observation</u> If A, B are simultaneously diagonalizable, then they commute:

<u>Proof</u> Say $S^{-1}AS = D$ = diagonal
$S^{-1}BS = E$ = diagonal.

Then

$$AB = (SD\underbrace{S^{-1})(S}E S^{-1}) = S\underbrace{DE}_{\text{commute!}}S^{-1}$$

$$= SEDS^{-1} = SES^{-1}SDS^{-1} = BA.$$

∎

<u>Theorem</u> Let $A, B \in M_n$ be diagonalizable. Then $AB = BA$ if and only if A, B are simultaneously diagonalizable.

<u>Proof</u>

($\Leftarrow$) Observation above ✓

($\Rightarrow$) Assume $AB = BA$ and WLOG assume that A is already diagonal (easy to see $AB = BA$ iff $CA = AC$ for every $C \sim A$). In fact, WLOG we can assume that

$$A = \begin{bmatrix} \lambda_1 I_{m_1} & & & 0 \\ & \lambda_2 I_{m_2} & & \\ & & \ddots & \\ 0 & & & \lambda_k I_{m_k} \end{bmatrix}$$

where $\lambda_1, \lambda_2, \dots, \lambda_k$ are the distinct eigenvalues of A repeated $m_1, m_2, \dots, m_k$ times respectively ($m_1 + m_2 + \dots + m_k = n$)

(otherwise, work with $P^T A P$ instead of the diagonal A, where P is a permutation matrix ($P^{-1} = P^T$))

Notice now that "$AB = BA$" has important implications:

AB is like B with rows scaled by diag. entries of A

BA is like B with columns scaled by diag. entries of A

So

$$AB = BA \Rightarrow \lambda_i b_{ij} = b_{ij} \lambda_j$$

$$\Rightarrow (\lambda_i - \lambda_j) b_{ij} = 0 \qquad (*)$$

So if i, j come from different grouping in the diagonal blocks of A, then

$$(*) \Rightarrow b_{ij} = 0$$

This means

$$B = \begin{bmatrix} \boxed{B_1} & & & \\ & \boxed{B_2} & & 0 \\ & 0 & \ddots & \\ & & & \boxed{B_k} \end{bmatrix}$$

where B_j is an $m_j \times m_j$ block

By lemma in the beginning, $\exists\, T_j$ invertible in M_{m_j} such that

$$T_j^{-1} B_j T_j = \text{diagonal}$$

Take now

$$T = T_1 \oplus T_2 \oplus \cdots \oplus T_k$$

$$= \begin{bmatrix} T_1 & & & 0 \\ & T_2 & & \\ & & \ddots & \\ 0 & & & T_k \end{bmatrix}$$

and observe that

$$T^{-1} B T = \begin{bmatrix} \boxed{T_1^{-1} B_1 T_1} & & 0 \\ & \ddots & \\ 0 & & \boxed{T_k^{-1} B T_k} \end{bmatrix}$$

= direct sum of diagonal matrices

= diagonal

i.e. A, D are simultaneously diagonalizable

since

$$T^{-1} A T = \begin{bmatrix} T_1^{-1} \lambda_1 I_{m_1} T_1 & & \\ & \ddots & \\ & & T_k^{-1} \lambda_k I_{m_k} T_k \end{bmatrix}$$

$= A =$ diagonal. ∎

<u>Experiment</u> 1) Find two commuting non-diagonalizable matrices A, B

2) Find two commuting matrices A, B such that one is diagonalizable and the other isn't.

(if possible)

A bit on <u>permutation matrices</u>

<u>Def</u> $P \in M_n$ is a <u>permutation</u> matrix if all its entries are in $\{0, 1\}$ & every column & every row has exactly one 1

Equivalently, ~~P has its columns for~~ the columns of P are $\{e_1, e_2, \ldots, e_n\}$ in some order, where $e_1, e_2, \ldots, e_n$ are the n columns of I.

- PA permutes the rows of A
- AP —— " —— column of A
- $P^T = P^{-1}$ $\left(P^T P = I \right)$ "orthogonal matrix"

So $P^T A P$

is a similarity transformation that is called "permutational symmetric"